Fiscal Year:	FY 2024	**Task Last Updated:**	FY 10/23/2023
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Project Title:	Phase Transitions in Colloid-Polymer Mixtures in Microgravity		
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Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
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COI Name (Institution):	Khusid, Boris Ph.D. (New Jersey Institute of Technology) Kondic, Lou Ph.D. (New Jersey Institute of Technology) Meyer, William Ph.D. (Universities Space Research Association/NASA Glenn Research Center)		
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Task Description:	Adding non-adsorbing polymers to a suspension of colloidal spheres induces attractive forces between spheres originating from the presence of depletion zones around the spheres. We propose to work on analyzing the data on colloid/non-adsorbing polymer mixtures acquired in microgravity in NASA-supported projects Binary Colloidal Alloy Test (BCAT) 3-6. In our recently completed project (Structure evolution during phase separation in colloids under microgravity, NNX16AQ79G, PI: Kondic, 8/16/2016 - 8/15/2018) we analyzed the data on suspensions of hard spheres collected in Physics of Hard Spheres (PHaSE) experiments. As part of that project, we have formulated a new model that includes up-to-date information about the nature of phase transitions in suspensions of hard spheres. This model, coupled with the data acquired in the NASA-supported projects, already allowed us to explain some important observations of the crystallization kinetics in suspensions of hard spheres. These results were presented at the 2017 and 2018 American Society for Gravitational and Space Research (ASGSR) meetings. As a part of the proposed work, we plan to develop a new model for phase transitions in colloid-polymer mixtures. The model that we propose to develop will be based on the one that we developed for suspensions of hard-spheres, but will involve significantly more complicated physics. The predictions of the model will be compared to the experimental data residing in the NASA Physical Sciences Informatics (PSI) system, as well as to relevant Earth-based experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Colloids have been studied extensively because of the many similarities they share with behaviors of atoms in a microscopic setting. The fact that the colloidal particles are macroscopic makes colloids much easier to observe than atomic systems, and yet their behavior is such that they may be viewed as "big atoms." The addition of polymer into a colloidal mixture leads to the possibility of three-phase coexistence. This project thus has the potential to give us new fundamental insights into the phase transition kinetics of systems with complex phase behavior. Experiments with colloids have applications in fields such as materials science, engineering, and pharmaceuticals. Understanding phase transitions on an atomic scale is extremely useful in making and working with materials that are used to manufacture electronics and other products that we use in our everyday lives.
Task Progress:	Colloidal particles can be considered as blocks for building materials, like atoms for forming molecules and crystals. The recent surge of activity in assembly of colloidal crystals is motivated not only by scientific curiosity but also by myriad technologies in materials engineering, electronics, life science, and chemical and pharmaceutical industries. Colloidal systems undergo phase transformations analogous to transformations in atomic and molecular systems. Therefore, they are considered as excellent model systems for the study of phase transitions since the particle motions can be observed by optical video microscopy and tracked by image processing. However, the rich variety of colloidal structures observed in numerous terrestrial experiments could also have been influenced by gravity effects, such as particle sedimentation, convection and jamming, which compete with colloidal forces during the slowly evolving structure formation. In this research program, we first process NASA Physical Sciences Informatics (PSI) video microscopy images to reveal the growth kinetics of colloid-polymer mixtures. We next conduct computer simulations of phase transitions in colloid-polymer mixtures under the conditions of microgravity experiments. Comparison of simulation results with data acquired from images recorded in microgravity experiments for over 5 orders of magnitude in time and more than 3 orders of magnitude in domain size will provide, for the first time, the unique opportunity to rigorously test theoretical predictions for the kinetics of phase transitions. Analysis of these images will be used to develop new coupled models, combining phase-field models for the initial stage of phase separation and crystallization models for the development of larger structures. The research results will provide recommendations for further development of theories, simulations, and ground-based experiments on phase transformations in advanced materials. We have processed NASA PSI video microscopy images to reveal the growth kinetics of colloid-polymer mixtures. We used the free software ImageJ to remove the extraneous background, adjust the brightness/contrast, and smooth the image, yielding a time-sequence of enhanced images that can be characterized quantitatively. We used the autocorrelation function to describe the time evolution of the length scale separating colloid-rich domains in the BCAT 3--5 experiments, and observed linear growth of domains with time. We have also developed a theoretical model for simulating phase separation in colloid-polymer systems. We are using the Cahn-Hilliard-Stokes equation for the dynamics of the colloid volume fraction and the associated fluid velocity. Both are assumed to be functions of space in two dimensions and time. The dynamics of the colloid volume fraction is driven by the chemical potential, which is derived from a free energy functional. We assume that the free energy can be split into two parts: one corresponding to the colloid, and the other to the polymer, the latter modeling the weakly attractive interactions between colloids. The free energy also contains a gradient-term that accounts for the diffuse nature of the interface between phases, as is typical in phase-field models. The velocity field is assumed to be determined by the incompressible Stokes equation forced by gradients in the colloid volume fraction through the so-called Korteweg stress. Our model differs from others in that the mobility is concentration-dependent; that is, the mobility decreases as the colloid volume fraction increases, and goes to zero as the crystal structure is approached. We expect this feature to significantly influence the simulated phase separation dynamics. Our model also differs from those previously proposed in that it accounts for viscosity variations in the Stokes equation; specifically, the viscosity is assumed to rise sharply with the colloid concentration. The role of hydrodynamic interactions in colloid-polymer phase separation has been the subject of controversy in the literature, so our simulations are able to shed light on the associated open questions. We have written a code in MATLAB for simulating the governing equations. The code uses a pseudospectral method in space and a semi-implicit time-stepping scheme based on splitting the mobility function. We validated our code against the results of a linear stability analysis, and performed convergence studies both in the time step and number of spatial grid points. Moreover, we have developed efficient algorithms for solving the variable-viscosity Stokes equations, which involves solving a large system of linear equations that determine the fluid velocity at every time step. Numerical simulations of the governing equations show that the system does indeed exhibit phase separation, with relatively colloid-poor areas formed amongst a background of colloid-rich ones. The colloid-poor areas become more circular as time goes on, and the system coarsens, in that these areas merge with nearby ones. We find that such merger events are correlated with dramatic increases in the fluid speed, and are associated with short-lived quadrupolar vortex structures that surround the merging domains. We find that including hydrodynamics causes the domains to become

	more circular, while omitting hydrodynamics encourages the formation of more elongated structures. Moreover, including hydrodynamic interactions improves the agreement with experiment of the predicted rate of domain growth. Our current direction is to conduct parameter sweeps of the model system to determine how colloid volume fraction, hydrodynamic interactions, and polymer concentration affect the phase separation dynamics. We also plan to assess how the coexistence of a colloidal liquid phase with gas and crystal phases influences the kinetics of the system, and how the vanishing mobility of the crystal phase affects the growth rate and geometry of the structures formed. We are currently writing up these results for publication in a peer-reviewed journal. Progress on this project was hindered by the COVID-19 pandemic. We have therefore submitted a request for a one-year no-cost extension (NCE) for this grant. The one-year extension will be used to complete work such as running simulations of our phase field model, with and without hydrodynamic interactions, to assess the role of hydrodynamic interactions in mediating phase separation. We will finish our processing and analysis of the Binary Colloidal Alloy Test (BCAT) 3-6 experimental data. We will also complete our numerical simulations of the Cahn-Hilliard-Stokes phase-field model, and compare the numerical results with experiments.
Bibliography Type:	Description: (Last Updated: 03/24/2025)
Abstracts for Journals and Proceedings	Oza AU. "Phase-field modeling of colloid-polymer mixtures in microgravity." International Congress for Industrial and Applied Mathematics, Tokyo, Japan, August 21, 2023. Abstracts. International Congress for Industrial and Applied Mathematics, Tokyo, Japan, August 21-25, 2023. , Aug-2023
Dissertations and Theses	Barnes LG. "Fluid dynamics of interacting particles: bouncing droplets and colloid-polymer mixtures." Dissertation, New Jersey Institute of Technology, July 2023. , Jul-2023